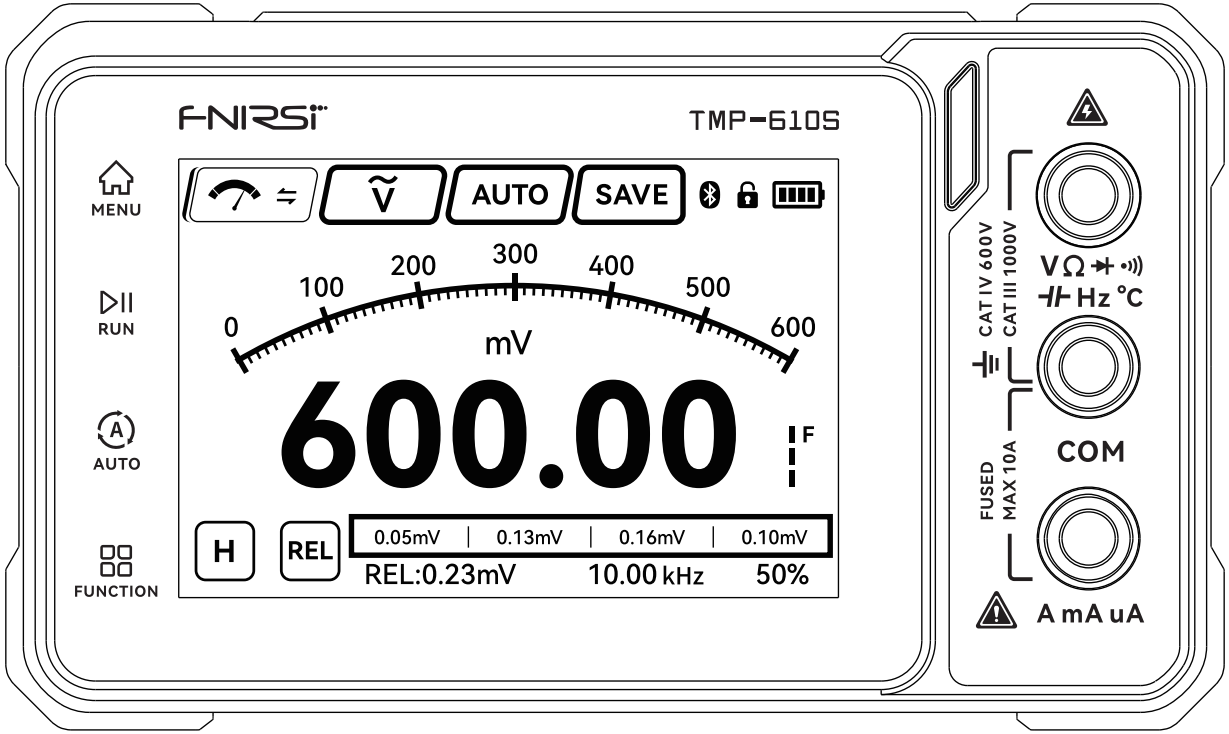
TMP-600/TMP-610S User Manual V1.0



※Please read this instruction manual carefully before using the product and keep it properly.

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1.Safety Statement

Please read this user manual carefully before using the product. The manual contains important safety information and should be kept properly for future reference. Users can visit our company website to check for updates to the user manual.

1.1 User Notice

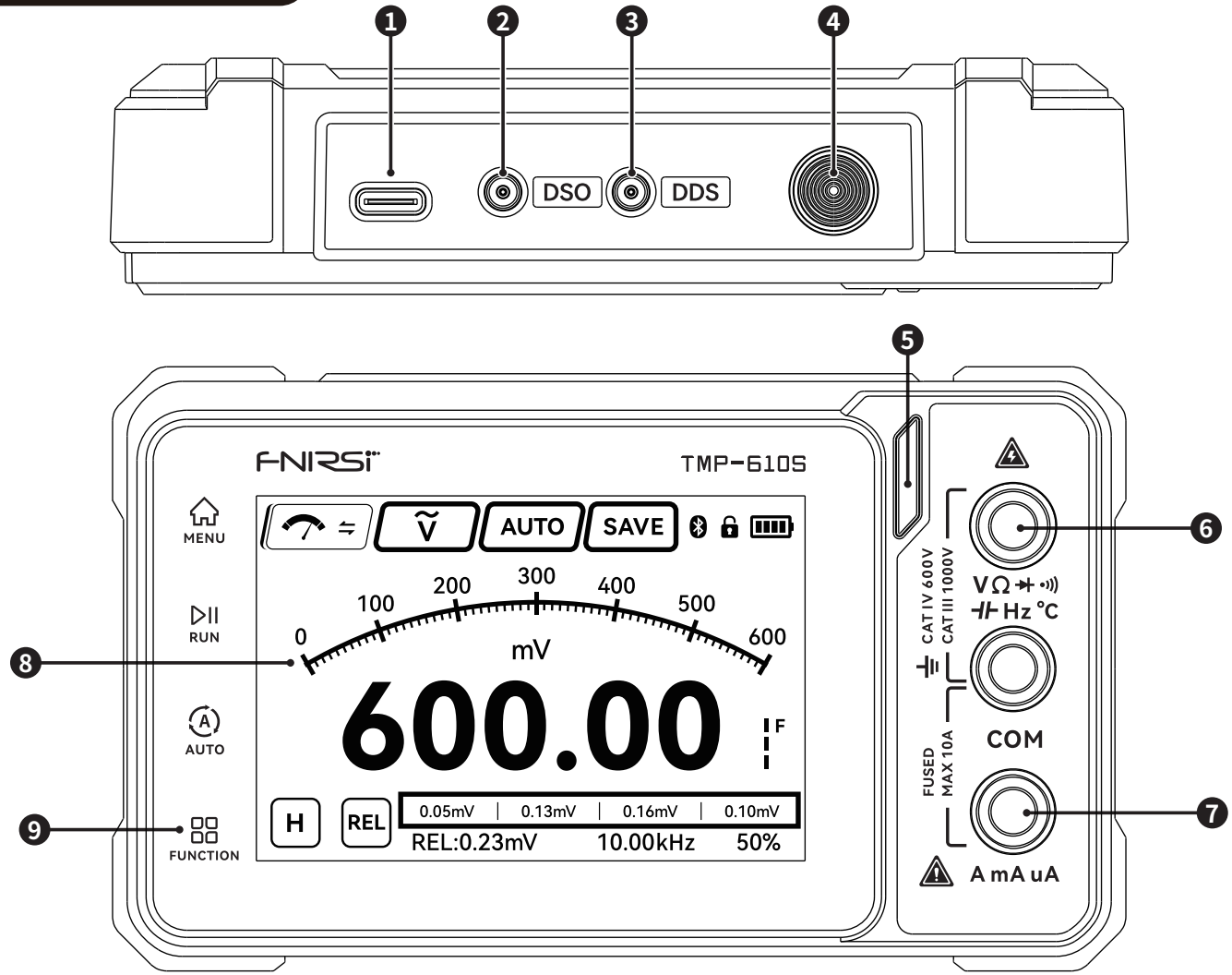
- Before using the TMP-600/TMP-610S, please carefully check the casing for any cracks or defects, and ensure the measurement ports and accessories are in normal condition.
- The TMP-600/TMP-610S is equipped with a rechargeable battery. For safety and measurement stability, do not perform measurements while charging. When the low battery warning appears, measurement accuracy may be affected; please recharge promptly.
- Please select the correct measurement ports, functions, and ranges to avoid damage to the instrument or personal injury.
- Hold the test leads or probes only by the insulated parts; do not hold the metal parts during measurement.
- Do not disassemble the casing arbitrarily, and do not operate measurements when the casing is open.
- Do not use the device in environments with flammable or explosive gases/materials, or in damp conditions.

2.Product Overview

2.1 Product Overview

The TMP-610S innovatively integrates three core functions: high-precision multimeter, oscilloscope, and multifunction signal generator. Its compact design meets the comprehensive measurement needs of engineers, researchers, and field technicians. The multimeter function features a 4½-digit display (maximum count 60,000), supports full-range auto-ranging and True RMS AC measurement for accurate reproduction of complex waveform energy, and offers 4 customized UI interfaces to suit different usage scenarios. With 50 MS/s sampling rate and 10 MHz bandwidth, the oscilloscope function enables precise capture of transient signal details. Additionally, the persistence feature dynamically overlays waveform traces, clearly displaying jitter, noise distribution, and abnormal events, providing strong support for in-depth fault analysis. The signal generator function can produce 13 standard waveform types with a maximum output frequency of up to 50 kHz.

2.2 Appearance Description



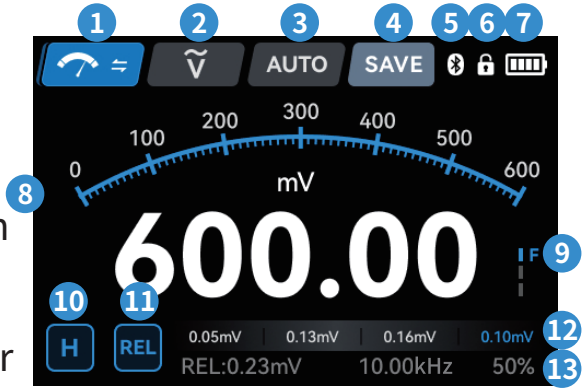
① Type-C port (for charging and USB communication)
② DSO Oscilloscope Interface
③ DDS Signal Generator Interface
④ Power button (long press to power on/off, double-tap to lock/unlock the screen)
⑤ Charging indicator light (red while charging, green when fully charged; works in both power-on and power-off states)
⑥ Voltage, Resistance, Capacitance, Continuity, Diode, Frequency, and Temperature Measurement Terminal (Black COM Terminal is the common ground)
⑦ Current Measurement Terminal (Black COM Terminal is the common ground)
⑧ 3.5-inch (480 × 320) touchscreen
⑨ Multifunction touch buttons

- Menu: opens the main menu
- Run: controls run/pause
- Auto: auto range (multimeter) / auto measurement (oscilloscope)
- FUNCTION: function settings

3.Quick Start

3.1 Description of the Multimeter Measurement Interface

Press【MENU】on the panel, select “Multimeter” to enter the measurement interface.



▶3.1.1 Normal Mode

①**Measurement Mode:** Tap the corresponding icon to switch between Normal Mode, Table Mode, Curve Mode, and Monitor Mode.

②**Measurement Range Area:** Select different measurement ranges for testing.

③**Range Indicator:** Supports Auto Range (AUTO) and Manual Range (available in Voltage, Current, Resistance, and Capacitance modes). AUTO+ range is also supported in Voltage and Current modes. In Voltage AUTO+ mode, the mV range is not available; in Current AUTO+ mode, the μA range is not available.

④**Data Save:** Tap “SAVE” to display the current multimeter reading in the Data Record Area.

⑤**Bluetooth Status:** The Bluetooth icon will not appear if Bluetooth is disabled in the settings. When enabled, a white Bluetooth icon is displayed. After connecting to the APP, the icon turns blue.

⑥**Touch Status:** Double-press the Power button to lock or unlock the touch operation of the device.

⑦**Battery Status:** Displays the current battery level when not charging. During charging, the charging progress is shown. (Measurements may be unreliable during charging, and there is a risk of leakage current.)

⑧**Main Display:** Displays the corresponding measurement value in real time according to the current measurement mode.

⑨**Sampling Rate:** Tap the measurement value area to switch between Fast, Medium, and Slow sampling speeds.

⑩**Data Hold:** Tap the corresponding icon to freeze the current measurement value. Tap again to release and resume real-time measurement display.

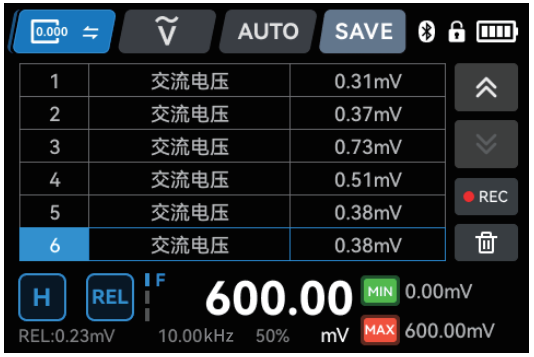
⑪**Relative Measurement:** Tap the corresponding icon to enable Relative Measurement and display the relative value. Tap again to cancel.

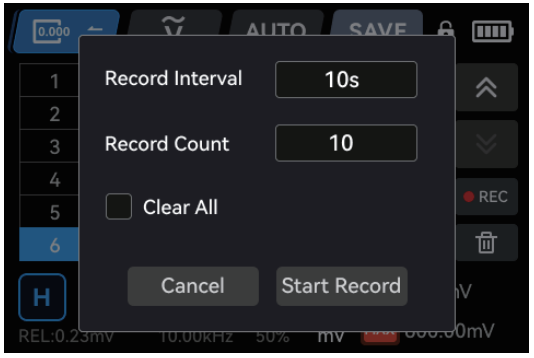
⑫**Data Record Area:** Displays saved data records, allowing users to view historical measurement data.

⑬**Frequency and Duty Cycle Display:** Available only in AC Voltage and AC Current modes.

▶3.1.2 Table Mode

- **REC Recording Function:** Tap the REC icon to enter the recording settings interface. When the device is in recording mode, the REC icon backlight turns red to indicate the current operating status.
- **Recording Interval:** The time interval for data recording can be customized to control the data acquisition frequency.
- **Records per Entry:** Sets the number of data records for a single recording session, with support for up to 100 records.
- **SAVE:** When REC recording is not enabled, tap “SAVE” to save the current measurement value to the table.



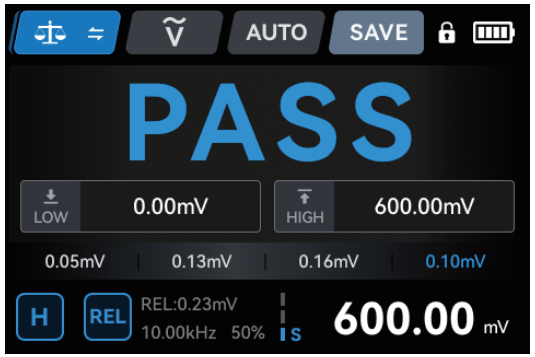


▶3.1.3 Curve Mode

- **Curve Timebase:** Used to adjust the waveform timebase. Select different settings according to measurement needs, including 160 ms/div, 320 ms/div, and 640 ms/div, allowing more detailed observation of waveform trends.

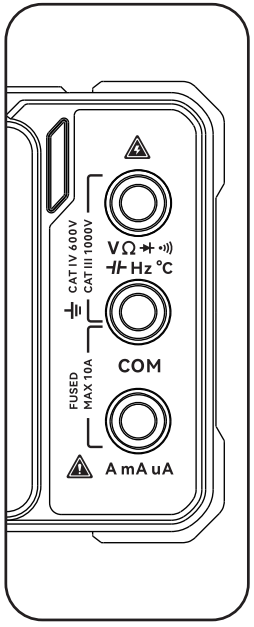


▶3.1.4 Monitor Mode



- Users can freely set the lower limit and upper limit in Monitor Mode.
- If the current value exceeds the preset range, “FAIL” will be displayed on the screen and the buzzer will sound continuously.
- If the current value is within the preset range, “PASS” will be displayed on the screen.

▶3.1.5 Multimeter Measurement Terminals Description



VΩ Hz °C Red Probe Terminal	1. AC/DC voltage measurement 2. Resistance measurement 3. Continuity test 4. Diode test 5. Capacitance measurement (discharge before measuring) 6. Frequency measurement 7. Temperature measurement
COM Black Probe Terminal	Common terminal for all measurements
A mA uA Red Probe Terminal	Dedicated Current Measurement Terminal, Max 10A (Do not use for other measurement modes) Built-in 10A Fuse: If blown, replace with a fuse of the same type.

▶3.1.6 Multimeter Specifications

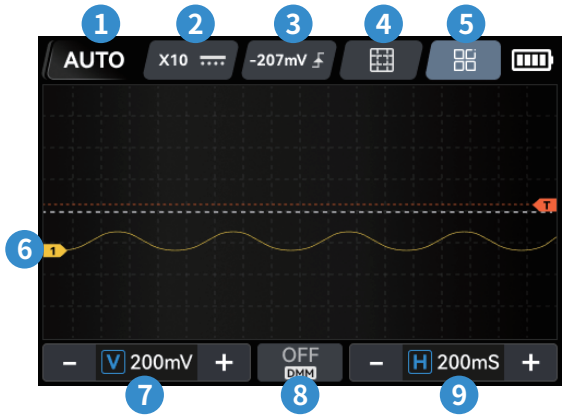
Multimeter Technical Specifications		
Function	**Range**	**Accuracy**
DC voltage	600mV/6V/60V/600V/1000V	± (0.03%+5d)
AC voltage	600mV/6V/60V /600V/750V	± (0.3%+15d)
DC current	600μA/6000μA/60mA/600mA	± (0.15%+5d)
	6A/10A	± (0.2%+5d)
AC current	600μA/6000μA/60mA/600mA	± (0.5%+15d)
	6A/10A	±(0.75%+15d)
Resistance	600Ω/6KΩ/60KΩ/600KΩ/6MΩ	± (0.1%+5d)
	60MΩ	± (0.2%+10d)
Capacitance	60nF/600nF/6μF/60μF	± (2.5%+30d)
	600μF/6mF/60mF	± (3.5%+30d)

Frequency	10Hz~60MHz	± (0.01%+5d)
Temperature	-40°C~1000°C (standard probe temperature measurement range: -20°C~300°C) -40°F~1832°F	± (1.0%+5°C) ± (1.5%+5°F)
Continuity	0Ω~1K Ω	± (0.2%+5d)
Diode	0V~3.0V, 0Ω~1K Ω	± (1%+5d)
AC Frequency Response	10KHz	/

3.2 Oscilloscope Measurement Interface Description

(Feature Available on TMP-610S)

Press the【MENU】button and select “Oscilloscope” to enter the measurement interface.



①**Trigger Status Display:** Shows the current trigger status in real time, including Trig'd, Auto, Ready, ROLL, STOP, Wait, allowing users to monitor the devices operating condition.

②**Channel Settings:** Allows settings such as probe attenuation ratio (X1, X10) and coupling type (DC, AC).

③**Trigger Settings:** Allows settings such as trigger mode (Auto, Normal, Single) and trigger edge (Rising, Falling, 50%).

④**Cursor Measurement:** Allows precise measurement of waveforms using the cursor function, supporting parameter positioning and value reading.

⑤**Function Settings:** Provides access to device configuration, allowing unified management of measurement and display parameters.

⑥**Channel Waveform:** Displays the raw waveform signal captured by the channel in real time.

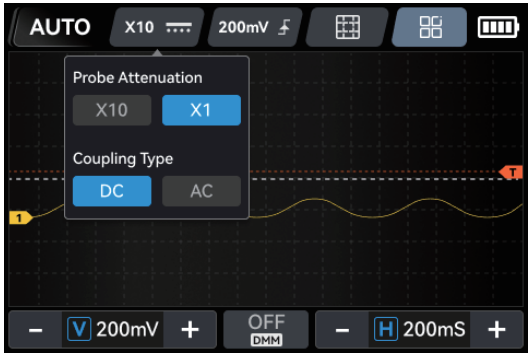
⑦**Vertical Scale Display:** Use the "–" and "+" keys to decrease or increase the vertical scale, adjusting the voltage scale in the vertical direction. Adjustable range: 10 mV~10 V.

⑧**Multimeter Function:** Supports one-touch enable/disable of the multimeter function; oscilloscope and multimeter functions can be used simultaneously.

⑨**Horizontal Timebase Display:** Use the "–" and "+" keys to decrease or increase the horizontal timebase, adjusting the time scale in the horizontal direction. Adjustable range: 50 ns~20 s.

▶3.2.1 Vertical System

(1) Channel Settings

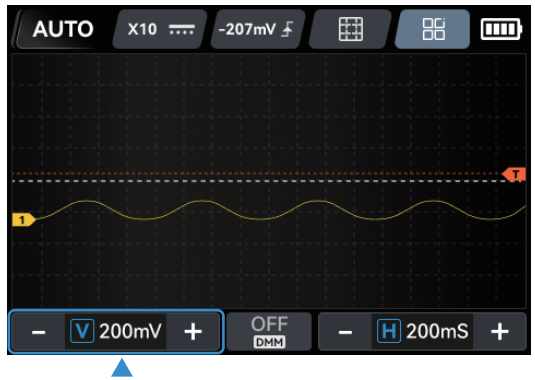


- Press the [Channel Settings] button, and the main interface will display options for "Probe Attenuation" and "Coupling Type".
- Select the appropriate probe attenuation (e.g., X1, X10) according to the signal voltage range.
- The user can choose different coupling modes as needed:

Coupling Mode	Description
DC	DC Coupling
AC	AC Coupling

(2) Vertical Scale

Vertical scale refers to the voltage value represented by each division in the vertical direction on the display. Use the “–” and “+” keys to decrease or increase the vertical scale, adjusting the voltage scale in the vertical direction. Adjustable range: 10 mV~10 V. At the same time, the scale information on the label at the top of the screen changes in real time. As shown in the right figure:

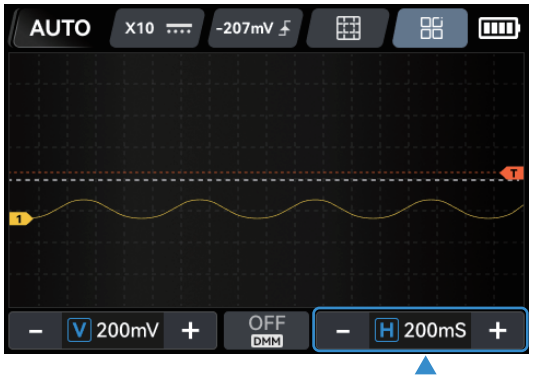


(3) Vertical Offset

Vertical offset refers to the displacement of the channel signal zero position of the waveform in the vertical direction relative to the center of the screen. When adjusting vertical offset, the corresponding channel waveform moves up or down. Click the vertical offset cursor to select vertical offset, then slide with one finger up or down in the waveform area to set the corresponding offset.

▶3.2.2 Horizontal System

Time scale refers to the time value represented by each division in the horizontal direction on the display. Use the "–" and "+" keys to decrease or increase the horizontal time base, adjusting the time scale in the horizontal direction. Adjustable range: 50 ns~20 s.
As shown in the right figure:



▶3.2.3 Trigger Settings

(1) Trigger Mode:

Affects the waveform length that the oscilloscope can display in continuous acquisition mode.

- Auto: The oscilloscope refreshes the waveform whether or not a trigger signal is detected.
- Normal: The oscilloscope refreshes the waveform only when a trigger signal is detected.

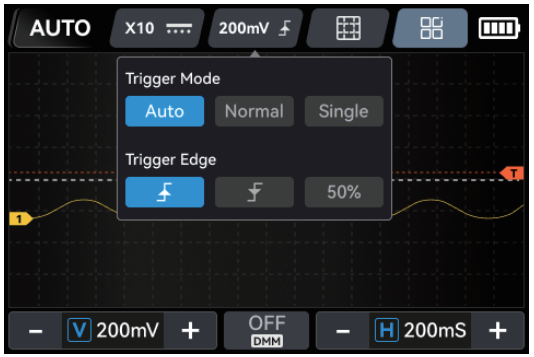
・Single: The oscilloscope stops data acquisition after the next trigger event. Used for capturing transient events.

(2) Trigger Edge:

・Rising Edge: The oscilloscope triggers when it detects a transition from low to high.

・Falling Edge: The oscilloscope triggers when it detects a transition from high to low.

・50%: Automatically sets the trigger level and timebase to the 50% position with one touch.

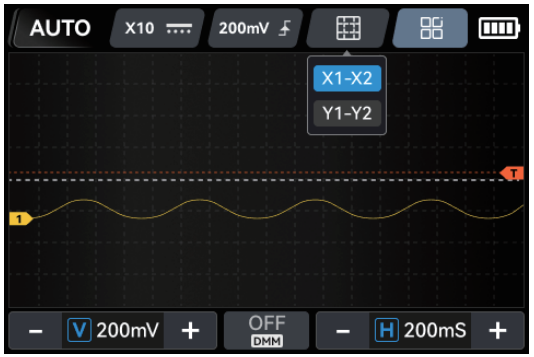


▶3.2.4 Cursor Measurement

Displays cursor lines on the waveform for measurement. Users can precisely locate specific points on the waveform using the cursors. When the cursor function is enabled, slide the screen to move the cursors on the waveform:

・Slide the screen to move Cursor 1/2 (X1–X2) left or right.

・Slide the screen to move Cursor 1/2 (Y1–Y2) up or down.



▶3.2.5 Function Settings

(1) Measurement Parameters

The oscilloscope's measurement function can accurately measure various parameters of the input signal, providing key data for signal analysis. Press the "FUNCTION" button on the panel, or tap the function option on the screen to operate. After operation, the oscilloscope will enter the measurement settings interface.



- All: Tap to display all measurement parameters on the main interface.
- Clear: Tap to remove all measurement parameters from the main interface.
- Frequency: Indicates the number of signal cycles per second.
- Period: Time for one complete signal cycle.
- +Duty (Positive Duty Cycle): Ratio of high-level duration to total period.
- –Duty (Negative Duty Cycle): Ratio of low-level duration to total period.
- +Width (Positive Pulse Width): Duration of the high-level signal.
- –Width (Negative Pulse Width): Duration of the low-level signal.
- Max: Maximum voltage of the signal.
- Min: Minimum voltage of the signal.

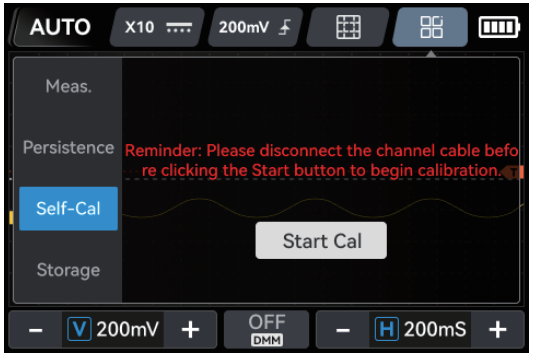
- P-P (Peak-to-Peak): Voltage between the highest and lowest points.
- Amplitude: Voltage difference between waveform top and bottom.
- RMS: Root mean square value of the signal.
- Average: Arithmetic average of the full waveform or the gated area.



(2) Persistence

The user can configure the persistence time.

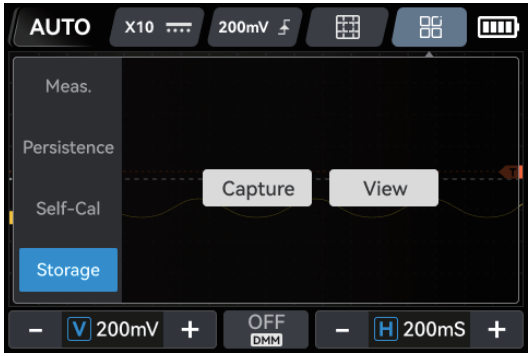
(3) Self Calibration



- Disconnect the channel cables to avoid external signal interference during calibration.
- Tap “Start Calibration” to initiate the internal calibration process. The oscilloscope will automatically calibrate key parameters such as vertical gain and horizontal time base.
- Upon successful calibration, the oscilloscope will automatically save the calibration data and notify the user that calibration is complete.

⑷ Storage

- Press the【Screenshot】on the screen; the oscilloscope will automatically exit the Function Settings interface and capture the screen.
- Select the【View】option to browse saved waveform images. Users can use the Select All, Edit, or Delete buttons to view or delete one or more images.



▶3.2.6 Oscilloscope Parameters

Parameter	Specification	Remarks
Real-time Sampling Rate	50MSa/s	/
Analog Bandwidth	10MHz	/
Input Impedance	1MΩ	/
Coupling Mode	AC/DC	/
Test Voltage Range	1:1 Probe: 80Vpp (±40V) /10:1 Probe: 800Vpp (±400V)	Oscilloscope set to X1 / X10

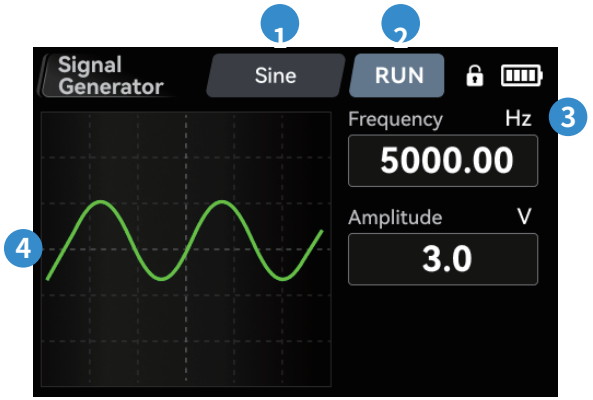
▶3.2.7 Oscilloscope Parameters

Parameter	Specification	Remarks
Vertical Sensitivity	10mV/div~10V/div	X1 range
Vertical Displacement	Adjustable with indication	/
Horizontal Timebase Range	50ns~20s	/
Trigger Mode	Auto, Normal, Single	/
Trigger Type	Rising, Falling, 50%	/
Trigger Level	Adjustable with indication	/
Measurement Parameters	Freq, Period, +Duty, –Duty, +Width, –Width, Max, Min, P–P, Amp, RMS, Avg	/
Persistence	Off / 500 ms / 1 s / ∞	/
Cursor Measurement	√	/
Waveform Screenshot Saving	√	/

3.3 Signal Generator Main Interface Description

(Feature Available on TMP-610S)

Press【MENU】and select “Signal Source” to enter the signal generator interface.



①**Output Waveform Selection:** The device supports 13 standard waveform outputs, including Sine, Square, Sawtooth, Half-wave, Full-wave, Positive Stair, Negative Stair, Exponential Rise, Exponential Fall, DC, Multi-tone, Sync Pulse, and Lorentz waveforms.

②**Run Status Toggle:** Tap the【RUN】on the panel or screen to quickly switch between Run and Stop states.

③**Waveform Parameter Adjustment:** Select the waveform parameter to adjust, then enter a value within the specified range in the pop-up settings interface to complete the adjustment.

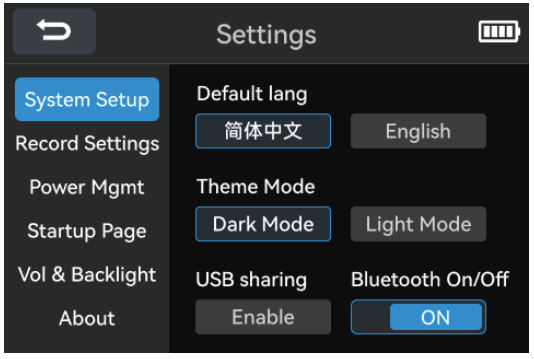
④**Waveform Illustration:** Provides a visual representation of the current output waveform.

▶3.3.1 Signal Generator Specifications

Parameter	Specification
Output Waveform	Supports 13 waveform types
Frequency	0~50 kHz
Amplitude	0.0V~3.0V
Duty Cycle	0~100%, adjustable for Square and Sawtooth waves

3.4 Other Settings

Press the【MENU】button on the panel and select “Settings” to enter the system settings interface.



Settings	Content	Description
Languages	Simplified Chinese, English	Currently supports 2 languages

Theme	Dark, Light	Switch device interface theme
USB Sharing	USB Connection	Used for data interactions, such as firmware updates and image uploads
Bluetooth Switch	OFF / ON	Turn Bluetooth On/Off
Recording Settings	Recording Interval	Custom recording interval to control the data acquisition frequency
	Records per Entry	Set the maximum number of data records per single recording, up to 100 records
	Clear All Data	Clear all saved historical measurement data with one touch
Power Management	Screen Timeout: 5min, 10min, 30min, Never	Dims to minimum brightness after timeout
	Power Off Time: 10min, 30min, 1h, Never	Auto power-off after set time
Startup Page	Multimeter, Oscilloscope, Signal Generator, Main Interface	Quick access to selected startup interface
Volume & Backlight	Buzzer: 0%–100%	0% disables the buzzer
	Backlight: 0%–100%	Min brightness: 0%
About	Restore Factory Settings	Restore Factory Settings
	Displays device basic info and manufacturer	/

3.5 Bluetooth Connectivity

① **APP Download:** Visit the official website (https://www.fnirsi.com/) to download and install the FNIRSI app.

② **Enter Settings:** Press the “MENU” key on the device panel, select “Settings”, and enter the System Settings interface.

③ **Enable Bluetooth:** In the System Settings interface, switch the Bluetooth function to “ON”.

④ **Device Connection:** Open the installed FNIRSI APP, perform a scan, and select the target device to complete pairing. Once connected, the Bluetooth status icon on the device will light up.

3.6 Firmware Upgrade

① **Enter Upgrade Mode:** With the device powered off, quickly press the Power key three times to enter the Firmware Upgrade interface.

② **Establish Connection:** Connect the device to a computer using a USB Type-C cable.

③ **Firmware Import:** The computer will automatically recognize the device as a USB drive. Drag the latest firmware file into the USB drive directory and wait for the transfer to complete.

④ **Complete Upgrade:** After the firmware transfer is finished, restart the device to complete the firmware upgrade process.

4. Troubleshooting

If no waveform appears on the screen after signal acquisition:

- Check if the probe is correctly connected to the signal cable.
- Check if the signal cable is properly connected to the DSO (channel connector).
- Check if the probe is properly connected to the device under test.
- Check if the device under test is generating a signal.
- Reacquire the signal.

If the measured voltage amplitude is larger or smaller than the actual value (note: this usually occurs when using probes):

- Check if the channel attenuation factor matches the probes actual attenuation ratio.

5. Maintenance

Cleaning the device exterior

- Regularly clean dust around the device and connectors to maintain good condition.
- Ensure no liquids, dust, or debris enter the device connectors.
- Dirty or damp jacks may affect measurement accuracy.

Check Battery and Power

- For oscilloscopes with built-in batteries, regularly check battery health. Avoid complete discharge; it is recommended to charge periodically and avoid long-term disuse.
- Use the official charger for charging to prevent overcharge or over-discharge, ensuring the battery stays within the proper operating voltage range.

Storage and Transport

- Storage environment: Store the instrument in a dry, well-ventilated place. Avoid high temperature, high humidity, or drastic temperature changes. Avoid direct sunlight exposure.
- Transport: Handle with care to prevent drops, especially during transport.

Software Updates

- Regularly check for new firmware updates. The latest firmware can fix known bugs and improve device performance.
- Follow correct procedures when updating, use official firmware files, and avoid power interruptions or other disturbances.

Restore Factory Settings

- If the device malfunctions or cannot operate normally, try restoring factory settings. This will erase all custom configurations and return the device to its initial state.
- Refer to the user manual or contact customer service for instructions on how to restore factory settings.

6.Contact Us

Product Name: 3 IN 1 Tablet Oscilloscope Multimeter (10 MHz)

Model: TMP-600/TMP-610S

Manufacturer: Shenzhen FNIRSI Technology Co., Ltd.

Address: 8th Floor, West Side, Building C, Weihua Da Industrial Park, Dalang Street, Longhua District, Shenzhen, Guangdong, China

Tel: 0755-28020752

Service Email: support@fnirsi.com

Business Email: business@fnirsi.com

Website: www.fnirsi.com

Standard Implemented: GB/T 13978-2008

http://www.fnirsi.com/

7.Warranty Information

※This page is the basic warranty card. Please keep it.

Thank you for choosing our company's products. The warranty period of this product starts from the date of sale. During the product warranty period, if the product is installed and used in accordance with the product manual and used in normal environment and conditions, and the fault is caused by defects in the original materials and processing, you can enjoy free repair services according to the content of this warranty clause. Please keep this warranty card properly as a warranty certificate. No reissue will be issued if it is lost.

The following situations will incur paid repair services:

- Unable to present the original valid warranty card.
- Damage caused by improper installation not meeting product requirements, standards, or relevant specifications.
- Damage caused by accessories in the installation environment not meeting product requirements, standards, or relevant specifications.
- Damage caused by improper use, improper storage, unauthorized disassembly, or unauthorized repairs by the user.
- Expiration of the warranty period.

Warranty Card

Product Model	TMP-600/TMP-610S	**Qty.**	
Distributor Name (where to buy)		**Contact**	
Address		**Invoice Number** (Order Number)	
Purchase Date (as per invoice)	**Year** **Month** **Day**		
User Name:		**Address:**	
Contact:		**Fault Description:**	



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